

FIITJEE
ALL INDIA TEST SERIES
FULL TEST – X
JEE (Main)-2022

TEST DATE: 10-04-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. $\tan \delta = \frac{V}{H} = \sqrt{3}$
 $\tan \delta' = \frac{V}{H'} = \frac{V}{H \cos \theta}$
 $\tan \delta' = \frac{\sqrt{3}}{\cos 30^\circ} = 2$
 $\delta' = \tan^{-1}(2)$

2. A

Sol. Since, all surfaces are frictionless
Acceleration $> g \sin \theta$

3. B

Sol. Truth table of AND gate

4. B

Sol. Voltage (regulated) across zener diode being 9 V, therefore ΔV across resistor is $(17 - 9) = 8V$

5. A

Sol. $mg = \left(\frac{\sigma^2}{2\epsilon_0} \right) A$; $q = \sigma A$
 $\Rightarrow mg = \frac{q^2}{2A\epsilon_0} \Rightarrow \epsilon = \sqrt{2mgA\epsilon_0}$

6. C

Sol. $N_0 e^{-\lambda t} = N_0 (1 - e^{-\lambda t})$

$$e^{-\lambda t} = \frac{1}{2}$$

$$t_{1/2} = \tau = \frac{\ln 2}{\lambda}$$

$$T = \frac{1}{\lambda} \ln 2$$

$$T = \tau$$

7. C

Sol. $I_A < I_B$ since COM is at $\frac{3R}{8}$ from I_A (close to axis A than to axis B)

8. D

Sol. Conceptual

9. C

Sol. $TP^{-1/3} = \text{constant}$
 $(PV/nR) \cdot P^{-1/3} = \text{constant}$
 $P^{2/3} V = \text{constant}$
 $PV^{3/2} = \text{constant}$

$$\Rightarrow x = \frac{3}{2}, \gamma = \frac{5}{3}$$

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - x} = \frac{3R}{2} + \frac{R}{1 - \frac{3}{2}} = -\frac{R}{2}$$

$\therefore C$ is negative

$$\Delta Q = nC\Delta T$$

$\Rightarrow$ if ΔQ is +ve ΔT is -ve

10. D

Sol. From the figure for real image formation

$$x + x' + 2f \geq 4f$$

$$\Rightarrow x + x' \geq 2f$$

11. C

Sol. Buoyant force on a bubble (B) = $\frac{4}{3}\pi r^3 \sigma g \uparrow$

$$\text{Viscous force (F)} = 6\pi\eta r v \downarrow$$

For uniform velocity $F = B$

$$\text{Equating both, we get } \eta = \frac{2r^2 \sigma g}{9v}$$

Putting values $\eta \approx 11$ poise

12. A

Sol. As long as cylinder is completely inside, B is constant in magnitude, then its value decreases linearly with displacement and finally becomes zero.

13. C

Sol. There is no change in fringe pattern, with symmetric motion of source. Intensity at maxima increases because intensity at slits increases.

14. D

Sol. Velocity of particles at extreme position is zero and acceleration of particle at extreme position is maximum.

15. B

Sol. Time period of SHM is independent of position of the tunnel

16. C

Sol. conceptual

17. D

Sol. $FR = Mg(R^2 - (R - a)^2)^{1/2}$

$$F = Mg \left[1 - \left(\frac{R - a}{R} \right)^2 \right]^{1/2}$$

18. C

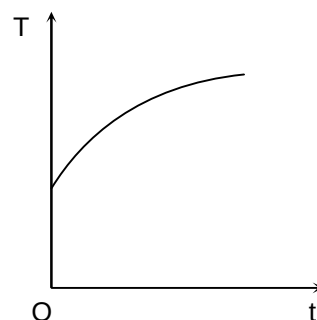
Sol. $\frac{dq}{dt} = \text{constant} = i$ (lets say)

$$i = aT^3 \frac{dT}{dt}, \quad \frac{dT}{dt} = \frac{i}{aT^3}$$

$$dT = \frac{i}{aT^3} dt$$

$$\int_{T_0}^T T^3 dT = \frac{i}{a} \int_0^t dt$$

$$\frac{T^4}{4} - \frac{T_0^4}{4} = \frac{i}{a} t$$



19. D

Sol. COM of $M(L, 0)$ and $M(-L, 0)$ is $(0, 0)$

Therefore COM of three particle system when third mass is at (x, y) is at $\left(\frac{x}{3}, \frac{y}{3}\right)$

COM traces path given by $x^2 + y^2 = \frac{L^2}{9}$

Where radius is $L/3$

20. B

Sol. $\vec{B} = \frac{\mu_0 q}{4\pi} \frac{(\vec{v} \times \vec{r})}{r^3}$

$$\vec{r} = a\hat{i} + b\hat{j}$$

$$\vec{v} = v\hat{i}$$

$$\Rightarrow \vec{B} = \frac{\mu_0 q}{4\pi} \frac{vb\hat{k}}{(a^2 + b^2)^{3/2}}$$

SECTION – B

21. 00200.00

 Sol. 1st overtone

$$f = \frac{2V}{2L} = \frac{V}{L} = \frac{1}{1} \sqrt{\frac{200}{(5/1000)}} = (200)$$

$$f = 200 \text{ Hz}$$

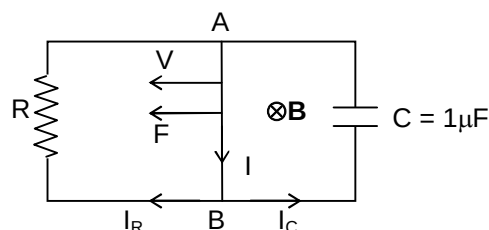
22. 00016.00

$$\text{Sol. } i = \frac{B\ell V}{R} + CB\ell \frac{dV}{dt}$$

$$i = \frac{B\ell}{R} \times \left[V + RC \frac{dV}{dt} \right]$$

$$m \frac{dV}{dt} = F - \frac{B^2 \ell^2}{R} \left(V + RC \frac{dV}{dt} \right)$$

$$\left[m + B^2 \ell^2 C \right] \frac{dV}{dt} = \left[F - \frac{B^2 \ell^2}{R} V \right]$$



When rod attains terminal velocity v_T , $\frac{dV}{dt} = 0$

$$v_T = \frac{FR}{B^2 \ell^2} = \frac{0.4 \times 10}{1 \times 0.25} = 16$$

23. 00021.00

$$\text{Sol. } \frac{1}{\lambda_{k_\alpha}} = R(z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \dots\dots (i)$$

$$\lambda_0 = \frac{hc}{eV_1}; \dots\dots (ii)$$

$$\left(\lambda_{k_\alpha} - \frac{\lambda_0}{2} \right) = \frac{22}{19} (\lambda_{k_\alpha} - \lambda_0)$$

$$\text{Therefore, } \lambda_{k_\alpha} = \frac{25}{6} \lambda_0;$$

Putting values from (i) and (ii) in above equation, we get $Z = 21$

24. 00027.00

Sol. Let say the displacement of cylinder from its equilibrium position is 'x'.

$$\Rightarrow ma = -\rho g A x$$

$$\Rightarrow a = - \left(\frac{\rho g A}{m} \right) x$$

$$\Rightarrow \omega_{\text{cylinder}} = \sqrt{\frac{\rho g A}{m}} \dots(i)$$

$$\Rightarrow \omega_{\text{spring-block}} = \sqrt{\frac{k}{m}} \dots(ii)$$

$$\Rightarrow \omega_{\text{cylinder}} = \omega_{\text{spring-block}} \dots(iii) \text{ (given)}$$

$$\Rightarrow K = \rho g A \text{ [from (i), (ii), (iii)]}$$

$$K = 27 \text{ N/m}$$

25. 00011.88

Sol. $\left(\frac{\beta}{1+\beta}\right)I_E = I_C$

$$\frac{99}{100} \times 12 = 11.88 \text{ mA}$$

26. 00060.00

Sol. $I_0 = 128 \text{ W/m}^2$; $I_1 = I_0/2$; $I_2 = I_0/2 \cos^2 \theta$
 $I_3 = I_0/2 \cos^4 \theta = 4$

27. 00002.00

Sol. Diameter = $2.4 + 2 \times LC - (0.4 + 2 \times LC) = 2 \text{ cm}$

28. 00050.00

Sol. $C' = \frac{\epsilon_0 \left(\frac{A}{2}\right)}{d} + \frac{2\epsilon_0 \left(\frac{A}{2}\right)}{d} = \frac{3}{2} \frac{\epsilon_0 A}{d} = 1.5 \text{ C}$

29. 00030.00

Sol. $2mg \sin \theta = mg \Rightarrow \sin \theta > \frac{1}{2}$
 $\theta = 30^\circ$

30. 00008.00

Sol. $A = A_0 e^{-bt}$

$$\frac{A_0}{2} = A_0 e^{-b(3)}$$

$$e^{-3b} = \frac{1}{2} \quad \dots (i)$$

$$A' = A_0 e^{-b(9)}$$

$$A' = A_0 \left(\frac{1}{8}\right) \text{ from equation (i)}$$

Chemistry

PART – B

SECTION – A

31. B

Sol. $\left(p + \frac{a}{V^2}\right)(V - b) = RT$

$$\frac{a}{V^2} = 0 \text{ (neglected)}$$

$$\therefore p(V - b) = RT$$

$$pV - pb = RT$$

$$\therefore pV = RT + pb$$

$$V = \left(\frac{R}{p}T + b\right)$$

Isobar is made at constant pressure between V(y-axis) and T(x-axis)

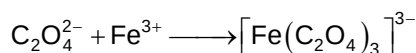
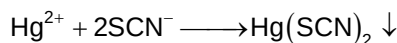
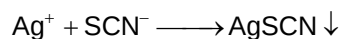
$$\text{Slope} = \frac{R}{p} = \tan \theta = 0.1642 \text{ dm}^3 \text{ K}^{-1}$$

$$\therefore p = \frac{R}{\tan \theta} = \frac{0.0821 \text{ dm}^3 \text{ mol}^{-1} \text{ K}^{-1}}{0.1642 \text{ dm}^3 \text{ K}^{-1}}$$

$$= 0.5 \text{ atm mol}^{-1}.$$

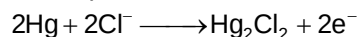
32. C

Sol. (A, B, D) colour disappears if a reactant added removes Fe^{3+} or SCN^- .



33. D

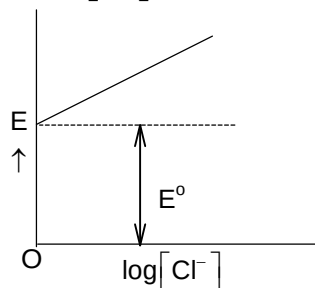
Sol. This represents oxidation half-cell



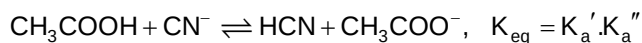
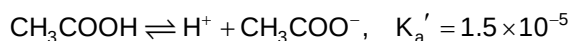
$$\therefore Q = \frac{1}{[\text{Cl}^-]^2} \Rightarrow E = E^\circ - \frac{0.0591}{2} \log Q$$

$$= E^\circ - \frac{0.0591}{2} \log \frac{1}{[\text{Cl}^-]^2} = E^\circ + 0.0591 \log [\text{Cl}^-]$$

When $[\text{Cl}^-] = 1 \text{ M}$, $E = E^\circ$ and E increases on increasing $\log [\text{Cl}^-]$



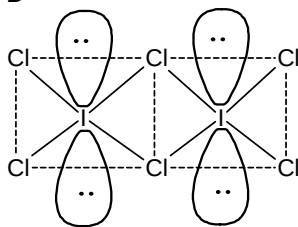
34. D



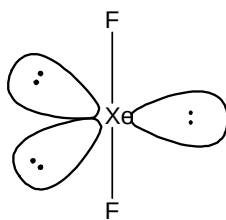
$$= 1.5 \times 10^{-5} \times \frac{1}{4.5 \times 10^{-10}} = 3.33 \times 10^4$$

35. D

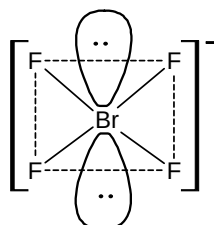
Sol.



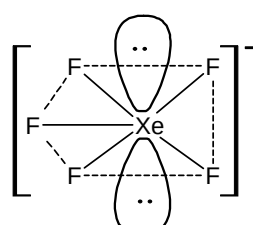
sp^3d^2
Planar



sp^3d
Linear: It is planar as three points are in same plane



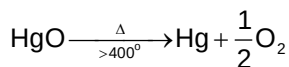
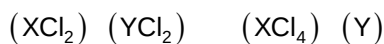
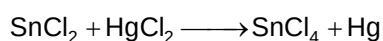
sp^3d^2
Square planar



sp^3d^3
Pentagonal planar

36. B

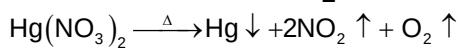
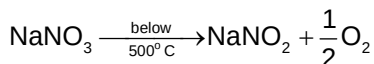
Sol.



HgS : Cinnabar

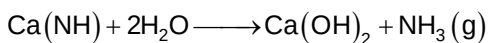
37. C

Sol.

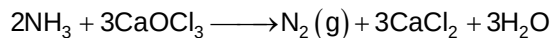


38. A

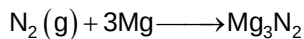
Sol.



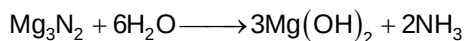
(B)



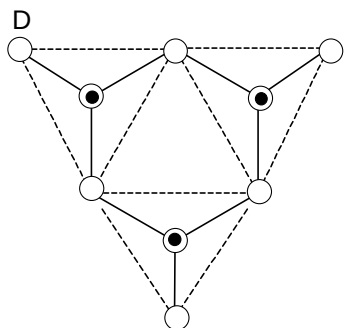
(B) (C)



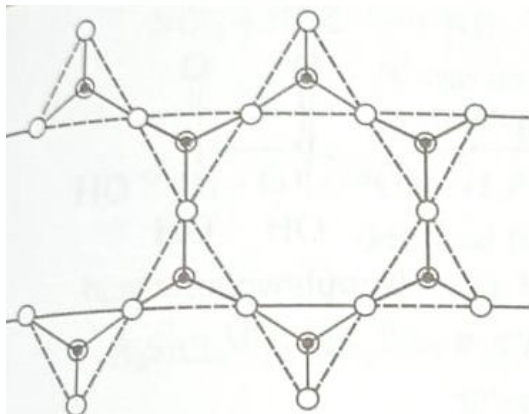
(C) (D)



(D) (B)

39.
Sol.


Cyclic silicate having three Si atoms ($\text{Si}_3\text{O}_9^{6-}$) contains three Si – O – Si linkage.



One tetrahedron shares three oxygen atoms while other share two oxygen atoms and these two tetrahedrons constitute one unit. Therefore, number of oxygen atoms shared per tetrahedron or

$$\text{Si atom} = \frac{3+2}{2} = 2\frac{1}{2}$$

⇒ General formula of double chain silicates : $(\text{Si}_4\text{O}_{11})_n^{6n-}$.

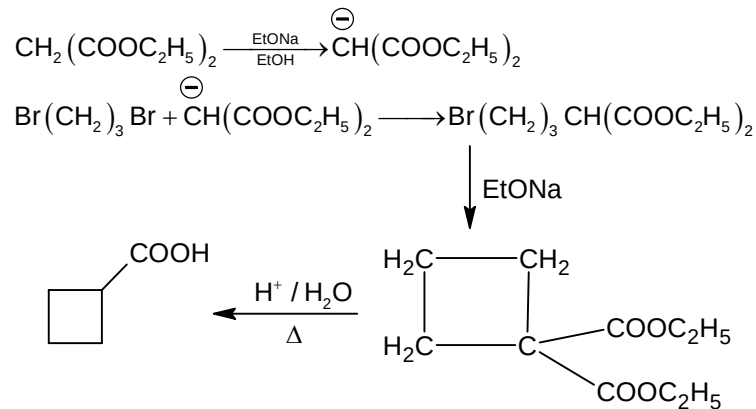
⇒ As the tendency of formation of multiple bond increases the extent of polymerization decreases. Silicon has large size hence, it forms less effective $p\pi - p\pi$ overlapping with oxygen.

40.

Sol. Due to presence of amide groups, nylon forms intermolecular H-bond with water and therefore, possesses high hydrophilicity.

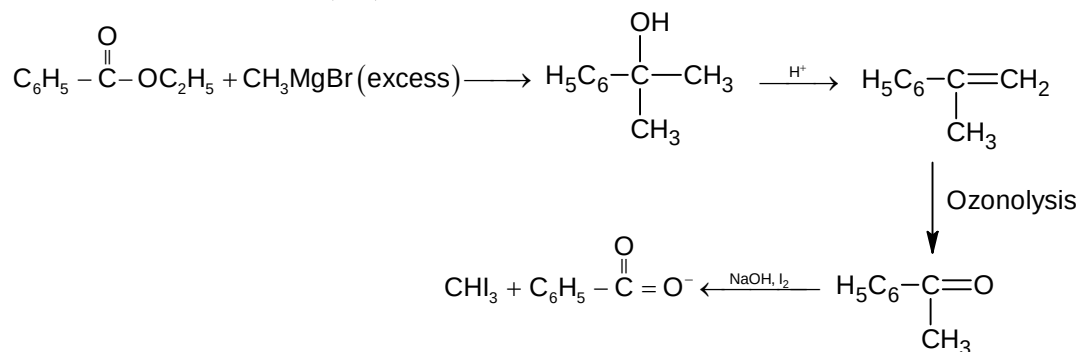
41.

Sol.



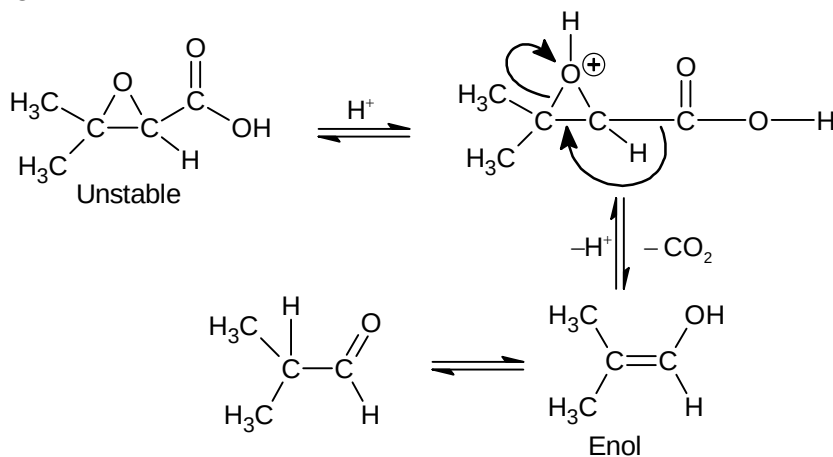
42. A

Sol. $C_9H_{10}O_2 + CH_3MgBr \xrightarrow{H^+} (B) \xrightarrow{\text{Ozonolysis}} \text{Ketone} \xrightarrow{\text{Positive iodoform test}}$
 $\begin{matrix} 10-5=5 \\ \text{(Olefin)} \end{matrix} \quad \begin{matrix} \text{(Excess)} \\ \text{(C}_8\text{H}_8\text{O)} \end{matrix}$



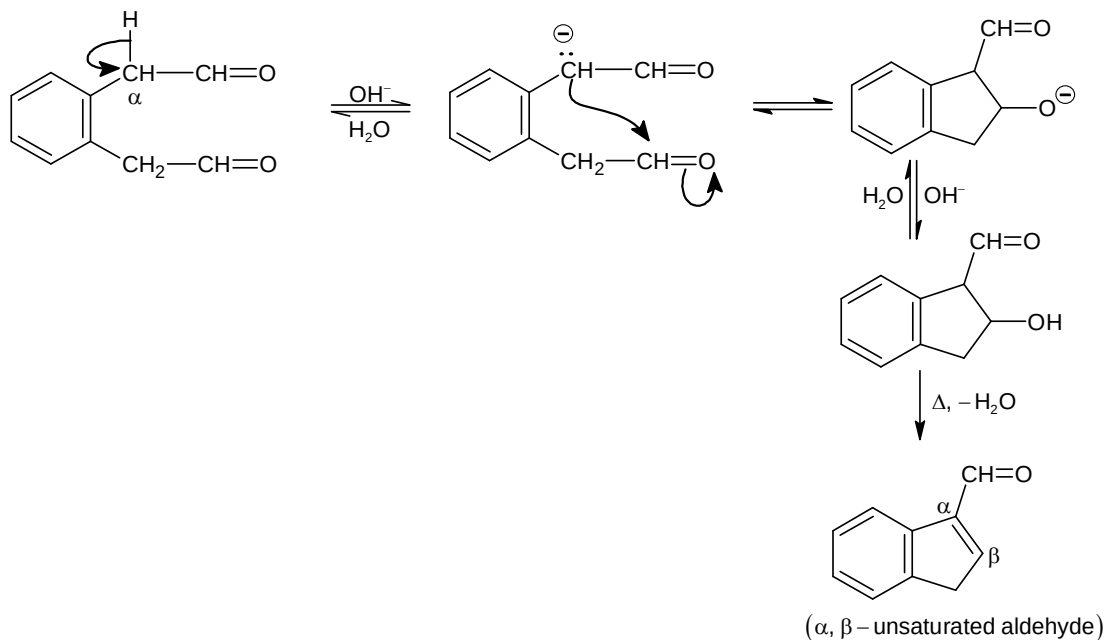
43. C

Sol.



44. B

Sol.



45. D

Sol. Glycine and all other amino acids are amphoteric because of the presence of NH_2 and CO_2H group both. The amino acid exists as Zwitterion and acid group is $-\text{NH}_3^+$ while basic group is $-\text{CO}_2^-$.

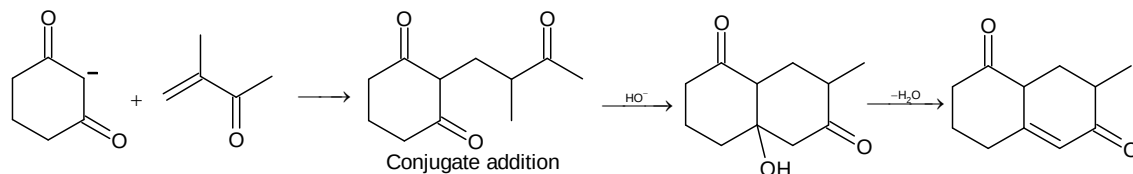
46. D

Sol. $\text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + 2\text{H}$

Oxidation of SO_2 and thus bleaching of coloured matter by reduction.

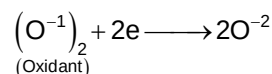
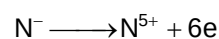
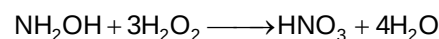
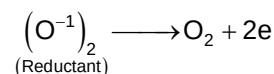
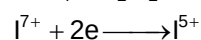
47. D

Sol.



48. A

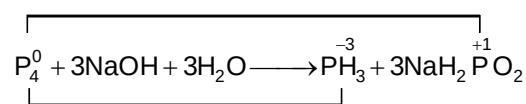
Sol. $\text{KIO}_4 + \text{H}_2\text{O}_2 \longrightarrow \text{KIO}_3 + \text{H}_2\text{O} + \text{O}_2$



49. C

Sol. The balanced disproportionation reaction involving white phosphorous with aq. NaOH is

Oxidation of P^0 to P^{+1} state



Reduction of P^0 to P^{-3} state

50. B

Sol. AgF is water soluble but CaF_2 is insoluble in water.

SECTION – B

51. 00707.20

Sol. We know that $\Delta T_t = K_t m$.

$$\text{Therefore, } m = \frac{0.0024}{1.86} = 0.0013 \text{ mol kg}^{-1}$$

or molarity = $0.0013 \text{ mol L}^{-1}$ (in case of solvent water molarity = molality).

The reaction involved is

	$I_2(aq)$	+	$I^- (aq)$	$\rightleftharpoons$	$I_3^- (aq)$
Initial moles	0.0492		0.1		0
Moles at equilibrium	$0.0492 - x$		$0.1 - x$		x

Total number of moles at equilibrium = $0.0492 - x + 0.1 - x + x = 0.1492 - x$.

Now, $0.1492 - x = 0.0013$ because of conc. of I_2 in all saturated solutions is same, so $x = 0.0479$.

Hence, the equilibrium constant is $K_c = \frac{[I_3^-]}{[I_2][I^-]} = \frac{0.0479}{0.0013 \times 0.0521} = 707.2$

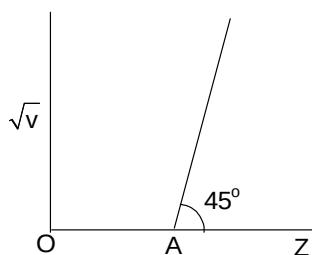
52. 00021.00

Sol. By Mosely's equation

$$\sqrt{v} = a(Z - b)$$

$$\sqrt{v} = aZ - ab$$

This represents a straight line between $\sqrt{v}$ (y-axis) and Z (x-axis)



At $\sqrt{v} = 0$, $aZ = ab$

$$\therefore Z = b = OA = 1$$

Also, slope = $a = \tan 45^\circ = 1$

When $\sqrt{v} = 400$, then $\sqrt{400} = Z - 1$

$$\therefore \sqrt{v} = Z - 1$$

$$\therefore Z = 21$$

53. 00005.00

Sol. Species with at least one or more unpaired electron(s) are attracted by the magnetic field and is said to be paramagnetic

Na^+ , Zn^{2+} , Cu^+ and O_2^{2-}

No unpaired electron hence, diamagnetic

O_2 – Two unpaired electrons, in anti-bonding molecular orbitals

Fe^{2+} – Four unpaired electrons in 3d-orbital

Cr^{3+} – Three unpaired electrons in 3d-orbital

Cu^+ – One unpaired electron in 3d-orbital

O_2^- – One unpaired electron in anti-bonding molecular orbitals

54. 00000.30

Sol. First vapour appears at 70 torr. It means total vapour pressure = 70.0 torr.

$$\therefore p_{\text{total}} = p_A^0 X_A + p_B^0 (1 - X_A)$$

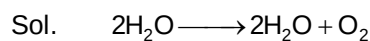
$$70.0 = 125X_A + 46.3(1 - X_A)$$

$$(70 - 46.3) = (125 - 46.3)X_A$$

$$23.7 = 78.7X_A$$

$$\therefore X_A = 0.30$$

55. 00009.00



$$k = 0.0693 \text{ min}^{-1}$$

$$\therefore T_{50} = \frac{0.693}{k} = \frac{0.693}{0.0693 \text{ min}^{-1}} = 10 \text{ min}$$

$$\text{H}_2\text{O}_2 \text{ left in 20 min, } y = \frac{20}{10} = 2$$

$$[A]_t = [A]_0 \left(\frac{1}{2}\right)^y = 1 \text{ mol} \left(\frac{1}{2}\right)^2 = \frac{1}{4} \text{ mol}$$

$$\text{Thus, H}_2\text{O}_2 \text{ decomposed} = 1 - \frac{1}{4} = \frac{3}{4} \text{ mol}$$

$$\text{O}_2 \text{ formed} = \frac{3}{8} \text{ mol}$$

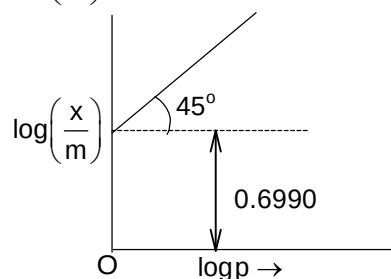
$$\therefore V = \frac{nRT}{p}$$

$$= \frac{\frac{3}{8} \text{ mol} \times 0.821 \text{ L mol}^{-1} \text{ K}^{-1} \times 293 \text{ K}}{\left(\frac{760}{760}\right) \text{ atm}} = 9.02 \text{ L}$$

56. 00004.00

 Sol. By Freundlich's adsorption isotherm $\left(\frac{x}{m}\right) = k(p)^{\frac{1}{n}}$

$$\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n} \log p$$



$$y = C + mx$$

$$\frac{1}{n} = m = \text{slope} = \tan 45^\circ = 1$$

$$\therefore n = 1$$

$$\log k = 0.6990 = C$$

$$\therefore K = 50$$

$$\frac{x}{m} = k(p)^{\frac{1}{n}} = 5.0(p)$$

$$\text{When, } p = 0.80$$

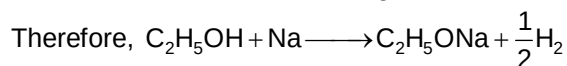
$$\frac{x}{m} = 5.0 \times 0.8 = 4$$

57.	00006.00	
Sol.	Complexes	Number of geometrical isomers
	(I) $[\text{Pt}(\text{NH}_3)_2(\text{SCN})_2]$	2
	(II) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$	2
	(III) $[\text{Pt}(\text{en})\text{Cl}_2]$	0
	(IV) $[\text{Cr}(\text{en})_2\text{Br}_2]^+$	2
	(V) $[\text{Rh}(\text{en})_3]^{3+}$	0
	(VI) $[\text{CoCl}_2\text{Br}_2]^{2-}$	0

Hence, sum of total number of geometrical isomers = 6.

58. 00007.48

Sol. Given, Hexene + Ethanol = 10 g



$$\text{Now, mole of H}_2 \text{ formed} = \frac{PV}{RT} = \frac{760 \times 200}{760 \times 1000 \times 0.0821 \times 300}$$

$$= 8.13 \times 10^{-3} \text{ mole}$$

($\because$ 200 ml H_2 is formed at 27°C and 760 mm pressure)

Now, $\frac{1}{2}$ mole H_2 is formed by = 46 g $\text{C}_2\text{H}_5\text{OH}$

$$8.13 \times 10^{-3} \text{ is formed} = 46 \times 8.13 \times 10^{-3} \times 2 \text{ g C}_2\text{H}_5\text{OH}$$

$$= 0.748 \text{ g}$$

$$\therefore \% \text{ of ethanol in mixture} = \frac{0.748 \times 100}{10} = 7.48\%$$

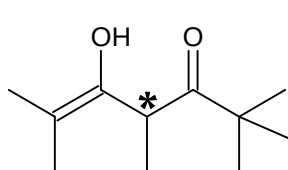
59. 00004.00

$$\text{Sol. Mass of O}_2 \text{ adsorbed} = \frac{PVM}{RT} = \frac{(10-2) \times 32}{400 \times 0.08} = 8\text{g}$$

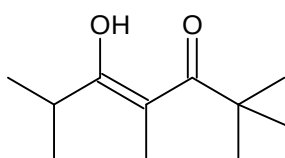
$$\therefore \frac{X}{m} = \frac{8}{2} = 4$$

60. 00006.00

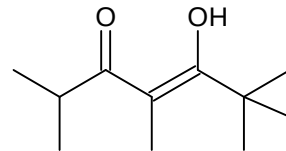
Sol.



(d,l)



(Two geometrical isomers)



(Two geometrical isomers)

Mathematics

PART – C

SECTION – A

61. C

Sol. Let $f(x) = x^3 + \cos 1 \cdot x^2 + \sin 1 \cdot x + 1 = 0$

$$f'(x) = 3x^2 + 2 \cdot \cos 1 \cdot x + \sin 1$$

$$f'(x) > 0$$

So, the real root is negative $\tan^{-1} \alpha + \tan^{-1} \frac{1}{\alpha} = -\frac{\pi}{2}$

62. A

Sol. Let $f(x) = (ax + b)(x - 2)(x - 3)(x - 4)$

$$f'(1) = 5a + 11b = 0 \Rightarrow -\frac{b}{a} = \frac{5}{11}$$

63. B

$$\text{Sol. } \frac{x^2 dy - y^2 dx}{x^2 y^2} = \frac{(xy) d(xy)}{x^2 y^2} \Rightarrow \int \frac{dy}{y^2} - \int \frac{dx}{x^2} = \int \frac{d(xy)}{xy} \Rightarrow -\frac{1}{y} + \frac{1}{x} = \log |xy| + c$$

64. B

Sol. $k^3 + 9k^2 + 26k + 24 = (k + 2)(k + 3)(k + 4)$

$$S = \sum_{k=0}^n \frac{(-1)^k (k+1)}{(k+1)(k+2)(k+3)(k+4)} \cdot {}^n C_k = \frac{1}{(n+1)(n+2)(n+3)(n+4)} \sum_{k=0}^n (k+1) \cdot {}^{n+4} C_{k+4}$$

$$\sum_{k=4}^{n+4} (-1)^k (k-3) \cdot {}^{n+4} C_k = \frac{(n+1)(n+2)}{2}$$

$$\text{So, } S = \frac{1}{2(n+3)(n+4)}$$

65. B

Sol. $|z| \leq 2$

$$\Rightarrow (a + \cos \theta)^2 + (2a - \sin \theta)^2 \leq 4 \Rightarrow 2a(\cos \theta - 2 \sin \theta) \leq 3 - 5a^2$$

$$\Rightarrow 2\sqrt{5} a \sin(\theta - \alpha) \leq 3 - 5a^2 \Rightarrow 2\sqrt{5} |a| \leq 3 - 5a^2$$

$$\Rightarrow a \in \left[-\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right]$$

66. D

Sol. For point of intersection $x - ax^2 = \frac{x^2}{a} \Rightarrow x = 0, \frac{a}{1+a^2}$

$$\text{Area} = \int_0^{\frac{a}{1+a^2}} \left(x - ax^2 - \frac{x^2}{a} \right) dx = \frac{5}{6} \frac{a^2}{(1+a^2)^2}$$

Area maximum when $a = 1$

67. B

Sol. Define A as the number of the elements in S, we have

$$A = \binom{2006}{1} 9^{2005} + \binom{2006}{3} 9^{2003} + \dots + \binom{2006}{2005} 9$$

On the other hand, $(9+1)^{2006} = \sum_{k=0}^{2006} \binom{2006}{k} 9^{2006-k}$ and $(9-1)^{2006} = \sum_{k=0}^{2006} \binom{2006}{k} (-1)^k 9^{2006-k}$

$$\text{So, } A = \binom{2006}{1} 9^{2005} + \binom{2006}{3} 9^{2003} + \dots + \binom{2006}{2005} 9 = \frac{1}{2} 10^{2006} - 8^{2006}$$

68. D

Sol. Since each has equally 9 different possible results for A and B to draw a ball from the packet independently, the total number of possible events is $9^2 = 81$. From $a - 2b + 10 > 0$ we get $2b < a + 10$. We find that when $b = 1, 2, 3, 4, 5$, a can take any value in $1, 2, 3, \dots, 9$ to make the inequality hold. Then we have $9 \times 5 = 45$ admissible events
 When $b = 6$, a can be $3, 4, \dots, 9$ and there are 7 admissible events
 When $b = 7$, a can be $5, 6, 7, 8, 9$ and there are 5 admissible events
 When $b = 8$, a can be $7, 8, 9$, and there are 3 admissible events
 When $b = 9$, a can only be 9, and there is 1 admissible events

$$\text{So, the required probability is } \frac{45 + 7 + 5 + 3 + 1}{81} = \frac{61}{81}$$

69. B

Sol.
$$L = \lim_{x \rightarrow 4} \left(1 + \frac{\log_5 \frac{5x}{4}}{\log_5 \frac{20}{x}} - 1 \right)^{\left(\log_5 \frac{x}{4} \right)^{-1}} \Rightarrow e^{\lim_{x \rightarrow 4} \frac{\log_5 \frac{x^2}{16}}{\log_5 \frac{x}{4}}} = e^2$$

 $\therefore [L] = 7$

70. B

Sol. In $\triangle ECD$, $\tan 3\alpha = \frac{h}{CD}$

$$\Rightarrow CD = h \cot 3\alpha \quad \dots (1)$$

In $\triangle EBD$, $\tan 2\alpha = \frac{h}{BD}$

$$\Rightarrow BD = h \cot 2\alpha \quad \dots (2)$$

In $\triangle EAD$, $\tan \alpha = \frac{h}{AD}$

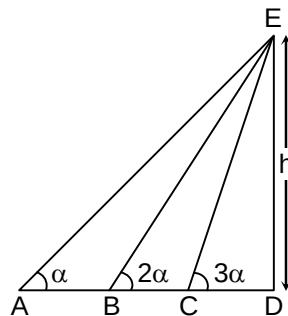
$$\Rightarrow AD = h \cot \alpha \quad \dots (3)$$

From equation (2) and (3), we have $AD - BD = h \cot \alpha - h \cot 2\alpha$

$$AB = h (\cot \alpha - \cot 2\alpha) \quad \dots (4)$$

From equation (1) and (4), we have $BD - CD = h \cot 2\alpha - h \cot 3\alpha$

$$\Rightarrow BC = h (\cot 2\alpha - \cot 3\alpha) \quad \dots (5)$$



From equation (4) and (5), we have
$$\frac{AB}{BC} = \frac{h(\cot \alpha - \cot 2\alpha)}{h(\cot 2\alpha - \cot 3\alpha)} = \frac{\frac{\cos \alpha}{\sin \alpha} - \frac{\cos 2\alpha}{\sin 2\alpha}}{\frac{\cos 2\alpha}{\sin 2\alpha} - \frac{\cos 3\alpha}{\sin 3\alpha}} = \frac{\frac{\sin(2\alpha - \alpha)}{\sin \alpha \sin 2\alpha}}{\frac{\sin(3\alpha - 2\alpha)}{\sin 2\alpha \sin 3\alpha}} = \frac{\sin 3\alpha}{\sin \alpha} = 3 - 4 \sin^2 \alpha = 3 - 2(1 - \cos 2\alpha) = 1 + 2 \cos 2\alpha$$

71. D

Sol. If variable is decreased by 1 then there is no effect on standard deviation

72. A

 Sol. $\sim(p \vee q) \vee (\sim p \wedge q) \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q) \equiv \sim p \wedge (\sim q \vee q) \equiv \sim p \wedge t \equiv \sim p$

73. D

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}; A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}; A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^{2n} = \begin{bmatrix} 1 & 0 & 0 \\ n & 1 & 0 \\ n & 0 & 1 \end{bmatrix}; n \in \mathbb{N}; A^{2n} - I = \begin{bmatrix} 0 & 0 & 0 \\ n & 0 & 0 \\ n & 0 & 0 \end{bmatrix}$$

74. D

 Sol. Equation of line parallel to z-axis passing through A is $\frac{x+1}{0} = \frac{y+2}{0} = \frac{z-4}{1} = \lambda$

 Point on the line is $(-1, -2, \lambda + 4)$ which lies on $x + y + z = 7$

$$-1 - 2 + \lambda + 4 = 7 \Rightarrow \lambda = 6$$

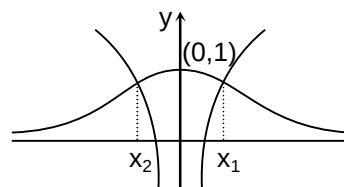
So, required distance is 6 units

75. B

 Sol. $\vec{r}_1 \cdot \vec{r}_2 = 0$

$$\ln |a|(1 + a^2) - a + a - 1 = 0$$

$$\ln |a| = \frac{1}{1 + a^2}$$

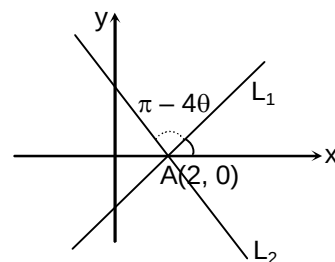
 From the graph it is clear that number of values of 'a' are two i.e., x_1, x_2 and $x_1 + x_2 = 0$


76. D

Sol. $\int_{-2}^x f(t)(x-t)dt + \int_x^2 f(t)(t-x)dt; p'(x) = \int_{-2}^x f(t)dt - \int_x^2 f(t)dt$

$$p''(x) = f(x) + f(x) = 2f(x); \int_0^2 \frac{p''(x)}{f(x)} dx = 2 \int_0^2 dx = 4$$

77. A

 Sol. Slope = $\tan\left(\frac{\pi}{2} - 2\theta + \theta\right) = \cot \theta = 2$


78. A

Sol. $\tan\left(\alpha - \frac{\pi}{4}\right) = \frac{\sin \frac{\pi}{2}}{3 - \cos \frac{\pi}{2}} = \frac{1}{3} \Rightarrow \frac{\tan \alpha - 1}{1 + \tan \alpha} = \frac{1}{3} \Rightarrow \tan \alpha = 2$

79. C

Sol. $\therefore$ Tangent at end of focal chord meet at directrix at 90°
 $\therefore$ PQ is focal chord and its mid-point will be circumcentre

Let $P = (at^2, 2at) \therefore Q = \left(\frac{a}{t^2}, -\frac{2a}{t}\right)$ and circumcentre = (h, k)

$\therefore 2h = a\left(t^2 + \frac{1}{t^2}\right)$ and $2k = 2a\left(t - \frac{1}{t}\right) \therefore 2h = a\left\{\left(t - \frac{1}{t}\right)^2 + 2\right\}$

$\therefore 2h = a\left\{\frac{k^2}{a^2} + 2\right\} \Rightarrow 2ha = k^2 + 2a^2 \therefore$ Locus is, $y^2 = 2a(x - a)$

$\therefore$ Focus = $\left(\frac{3a}{2}, 0\right)$

80. B

Sol. $\int \frac{(\sec^2 x - 1)\tan x}{(\tan^2 x + 2)} dx$; Put $\sec x \rightarrow t \Rightarrow \sec x \tan x dx = dt$
 $= \int \left(\frac{t}{1+t^2} - \frac{1}{t(t^2+1)}\right) dt \Rightarrow \log(1+t^2) - \log t + C$
 $\Rightarrow \log \left|\frac{1+\cos^2 x}{\cos x}\right| + C \Rightarrow \log \left|\frac{2-\sin^2 x}{\cos x}\right| + C$

SECTION - B

81. 09800.00

Sol. Number of integral points within the area above the x-axis is $\sum_{k=1}^{99} (99 - k) = 99 \times 49 = 4851$

The total number of integral points within the area is $2 \times 4851 + 98 = 9800$

82. 00002.00

Sol. $f(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$

So, $2f(x) = f(\sqrt{2}x) \Rightarrow f(x+a) \geq f(\sqrt{2}x)$

As $f(x)$ is increasing in R , then $x+a \geq \sqrt{2}x$

$\Rightarrow a \geq (\sqrt{2}-1)x$ since $x \in [a, a+2]$; $(\sqrt{2}-1)x$ is maximum at $x = a+2$

So, $a \geq (\sqrt{2}-1)(a+2)$; $a \in [\sqrt{2}, \infty)$

83. 00050.00

Sol. $2ab \leq a^2 + b^2$ using this

$S \leq \frac{\sin^2 x_1 + \cos^2 x_2}{2} + \dots + \frac{\sin^2 x_{100} + \cos^2 x_1}{2} = 50$

84. 03960.00

Sol. It is easy to see that there are $\binom{4}{2} \cdot P_2^4 = 72$ different ways to put two 'a's into the matrix such that each row and each column contains at most one 'a'. similarly, there are also $\binom{4}{2} \cdot P_2^4 = 72$ different ways for two 'b's to do the same thing. By the multiplicative principle, we get 72^2 ways. Among them we exclude 72 cases in which two 'b's occupy the same cells as the two 'a's do, and exclude $\binom{16}{1} \cdot P_2^9 = 16 \times 72$ cases in which one 'a' shares the same cell with a 'b'.
Therefore there are $72^2 - 72 - 16 \times 72 = 3960$ different ways that meet the requirement.

85. 00008.00

Sol. Let $z = \cos x + i \sin x$

$$\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = \frac{|z^2 + \bar{z}^2|}{|z|^2} = 2|\cos 2x| = 1$$

$$\cos 2x = 1 \quad \text{or} \quad \cos 2x = -1$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3} \text{ and } \frac{5\pi}{3}$$

So, 8 solutions

86. 00001.00

Sol. From $\frac{1}{a} + \frac{1}{b} \leq 2\sqrt{2}$, we have $a + b \leq 2\sqrt{2} ab$

$$\text{On the other hand } (a + b)^2 = 4ab + (a - b)^2 = 4ab + 4(ab)^3 \geq 4 \cdot 2\sqrt{ab \cdot (ab)^3} = 8(ab)^2$$

$$\text{And that means } a + b \geq 2\sqrt{2} ab \quad \dots (1)$$

$$\text{Therefore, } a + b = 2\sqrt{2} ab \quad \dots (2)$$

The equality in (1) holds only where $ab = 1$. Associating it with (2), we get

$$\begin{cases} a = \sqrt{2} - 1, \\ b = \sqrt{2} + 1, \end{cases} \text{ and } \begin{cases} a = \sqrt{2} + 1, \\ b = \sqrt{2} - 1 \end{cases}. \text{ So the answer is } \log_a b = -1$$

87. 00000.25

Sol. It is easy to find that $a = f(1) - f(0)$, $b = f(0)$, then

$$ab = f(0) \cdot (f(1) - f(0)) = -\left(f(0) - \frac{1}{2}f(1)\right)^2 + \frac{1}{4}(f(1))^2 \leq \frac{1}{4}(f(1))^2 \leq \frac{1}{2}$$

$$\text{When } 2f(0) = f(1) = \pm 1, \text{ i.e., } a = b = \pm \frac{1}{2}, \text{ we get } ab = \frac{1}{4}$$

$$\text{Therefore, the maximum of } ab \text{ is } \frac{1}{4} = 0.25$$

88. 00052.00

Sol. $N = a(111 \ 111)$

$924 = 4 \times 7 \times 11 \times 3$ so a must be divisible by 4, $a = 4, 8$

Sum of roots of equation = 11

If $a = 4$, then $b = 7, \lambda = 28$; $a = 8$, then $b = 3, \lambda = 24$

89. 00016.00

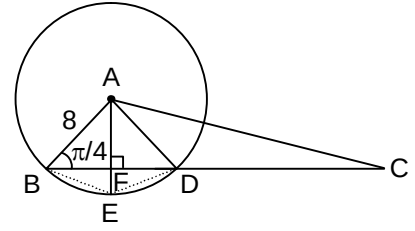
Sol. $AF = 4\sqrt{2}$

$$FE = 8 - 4\sqrt{2}$$

$$BD = 8\sqrt{2}$$

$$\text{Area of } \triangle BED = \frac{1}{2} \cdot 8\sqrt{2} (8 - 4\sqrt{2}) = 32(\sqrt{2} - 1)$$

$$\alpha = 32 ; \beta = 2$$



90. 00000.00

Sol. Let $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_{2022}$ be the vectors

$$\therefore \vec{a}_1 + \vec{a}_2 + \dots + \vec{a}_{2021} = \alpha \vec{a}_{2022}$$

$$\Rightarrow \vec{a}_1 + \vec{a}_2 + \dots + \vec{a}_{2020} + \vec{a}_{2022} = \beta \vec{a}_{2021}$$

$$\Rightarrow \alpha \vec{a}_{2022} - \beta \vec{a}_{2021} = \vec{a}_{2021} - \vec{a}_{2022}$$

$$\Rightarrow \alpha + 1 = \beta + 1 = 0$$

$$\Rightarrow \alpha = \beta = -1$$

$$\Rightarrow \vec{a}_1 - \vec{a}_2 + \vec{a}_3 + \dots + \vec{a}_{2022} = 0$$